Carlos Eduardo Bastos

Embedded Systems & Hardware Developer

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PROFESSIONAL SUMMARY

Embedded Systems & Hardware Developer with experience in industrial automation systems, including embedded Linux (Yocto), firmware validation, and PCB-based hardware platforms. Worked across product development and manufacturing environments, supporting software releases, hardware testing, and system-level integration. Background in microcontroller-based applications (ESP32, PIC) using Embedded C/C++, with hands-on experience in distributed systems using RS-485 communication. Open to international opportunities and relocation.

TECHNICAL SKILLS

- **Embedded:** Microcontrollers (ESP32, PIC), Embedded C/C++, Firmware Development
- **Hardware:** PCB Design, Circuit Analysis, Electronic Testing
- **Software:** C, C++, Python, Linux
- **Tools:** Git/GitHub, KiCad/Eagle, Qt, Vs Code

LANGUAGES

Portuguese (Native) | English (Full Professional Proficiency) | Spanish (Basic)

EXPERIENCE

Technical Support Analyst – Companytec Automação e Controle Ltda

Dec 2022 – Present

- Validated embedded software releases for industrial automation systems, testing an average of 2 firmware/backend versions per month during active product development
- Identified and reported software defects across backend and frontend layers, contributing directly to system stability and feature improvements
- Supported a Linux-based (Yocto) embedded platform responsible for full gas station automation, including pumps, tank monitoring, and environmental control
- Executed system-level validation prior to production deployment, ensuring reliable hardware/software integration

Production and Maintenance Technician – Companytec Automação e Controle Ltda

Dec 2020 – Dec 2022

- Programmed and operated two SMT production lines (Mirae MXP-100 and Panasonic AM100) supporting large-scale PCB assembly
- Performed board-level electrical and functional validation from PCB assembly through final product testing
- Contributed to continuous process improvement initiatives alongside engineering and manufacturing teams
- Supported full production flow, from PCB population to final industrial automation system testing

Embedded Systems & Hardware Developer (Voluntary) – Research & Innovation Lab

Feb 2016 – Nov 2020

- Designed and developed a distributed embedded monitoring and control system using RS-485 communication
- Implemented firmware for up to 32 networked nodes with real-time sensor data acquisition
- Designed and validated PCBs for distributed modules deployed in field demonstrations
- Presented the project at national science fairs and internationally in the United States (GENIUS Olympiad – Bronze Medal) and Argentina

- Mentored students in embedded systems development and hardware design practices

EDUCATION

Bachelor of Electrical Engineering - IFSUL

Electronics Technician - IFSUL

HONORS & AWARDS

Bronze Medal - Science Category, GENIUS Olympiad, 2018 (USA)

Gold Medal - Electronic Engineering Category, 32nd MOSTRATEC, 2017 (Brazil)

Brazil • Open to relocation • Embedded / Hardware Development